LECTURE-5

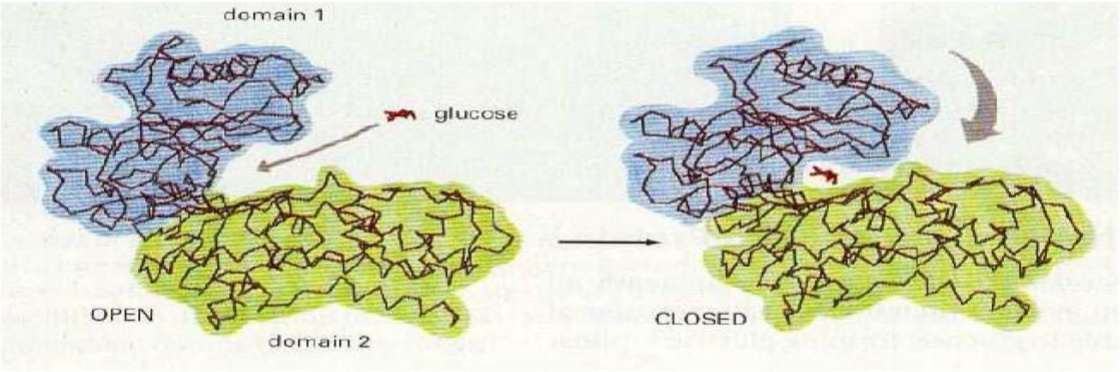
MM Kinetics
Tutorials

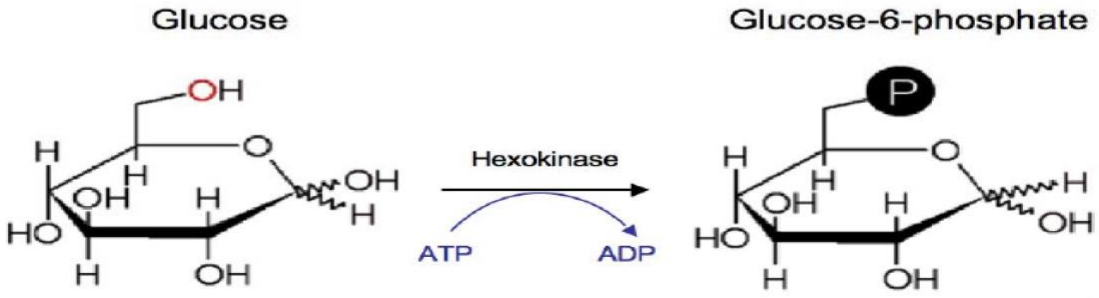
INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Michealis-Menten kinetics-Example

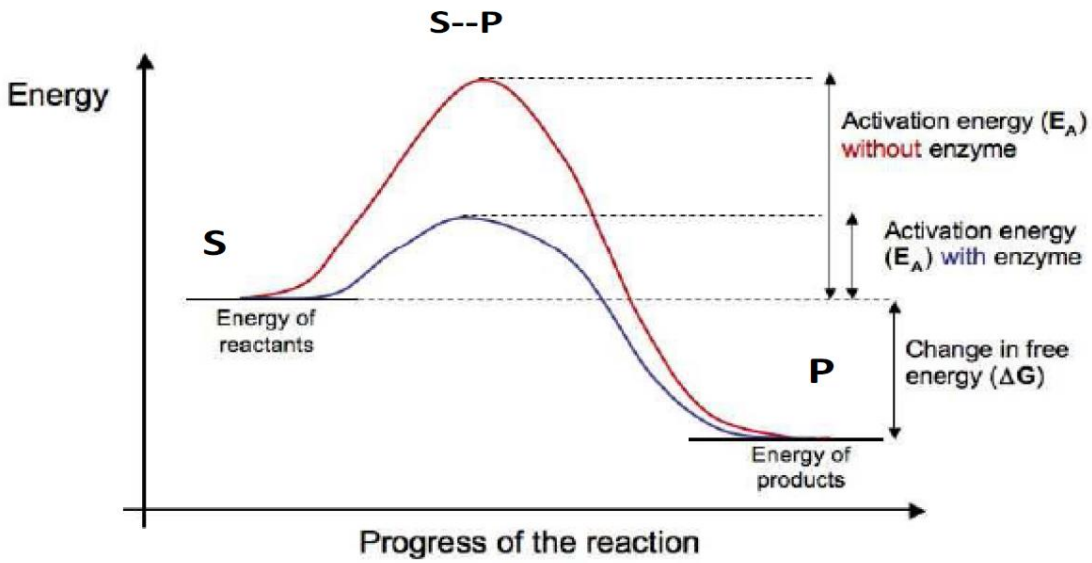
- An example of an enzymatic reaction is the first reaction of the glycolysis, catalysed by the enzyme hexokinase/

$$\text{Glucose} + \text{ATP} \rightarrow \text{Glucose-6-phosphate} + \text{ADP}$$





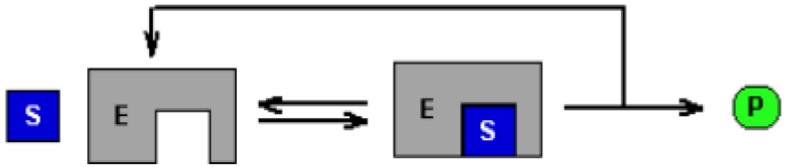
How enzyme catalyze the reaction?



- In the above reaction $S \underset{k_{-1}=10^{-5}}{\overset{k_1=10^{-3}}{\rightleftharpoons}} P and k_{eq} = \frac{[P]}{[S]} = 100.$
- If both k_1 and k_{-1} increase by 10^6 fold by enzyme, still K_{eq} will be

$$k_{eq} = \frac{10^{(-3+6)}}{10^{(-5+6)}} = 100.$$

Enzyme catalysis – Derivation of Michealis–Menten equation

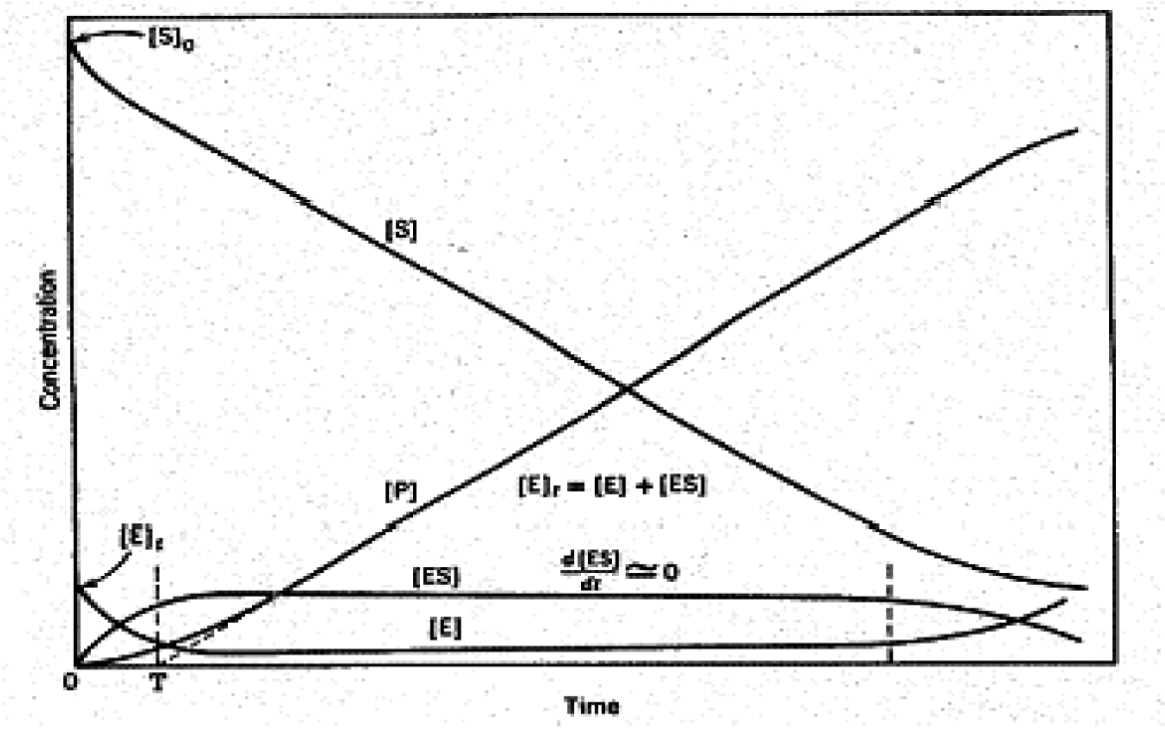


- The reaction scheme can be written as follows (C=complex between E and S, namely [ES]):
- $$E + S \underset{k_{-1}}{\overset{k_1}{\longleftrightarrow}} C \xrightarrow{k_2} E + P$$
- The equation for the different species follow the mass action and the evolution law is given as:

Derivation of Michealis-Menten equation

$$
\begin{aligned}
\frac{d[S]}{dt} &= -k_1[E][S] + k_{-1}[C] \\
\frac{d[E]}{dt} &= -k_1[E][S] + k_{-1}[C] + k_2[C] \\
\frac{d[C]}{dt} &= k_1[E][S] - k_{-1}C - k_2[C] \\
\frac{d[P]}{dt} &= k_2[C]
\end{aligned}
$$

Quasi-steady state approximation



- In this $\frac{d[C]}{dt} = \frac{d[E]}{dt} = 0$
- We define E_{tot} the total concentration of enzyme: $E_{tot} = E + C =$ constant

Quasi-steady state approximation

$$\frac{d[S]}{dt} = -k_1[E][S] + k_{-1}[C]$$
$$\frac{d[E]}{dt} = -k_1[E][S] + k_{-1}[C] + k_2[C] = 0$$
$$\frac{d[C]}{dt} = k_1[E][S] - k_{-1}C - k_2[C] = 0$$
$$\frac{d[P]}{dt} = k_2[C]$$

- The rate of change of total enzyme and complex is

$$\boxed{\frac{d[E]}{dt} = 0, \frac{d[C]}{dt} = 0}$$

- From this equation, $E_{tot} = E + C$, we can extract C, and its given as

$$\boxed{C = \frac{k_1 E_{tot}[S]}{k_1[S] + (k_1 + k2)} = \frac{E_{tot}[S]}{[S] + \frac{(k_1+k2)}{k_1}}}$$

Final equation

- When we replace this expression for C in the rate of appearance of P, we obtain:
  $v = \frac{d[P]}{dt} = k_2[C] = \frac{k_2 E_{tot}[S]}{[S]+\frac{(k_1+k2)}{k_1}}$
- This is usually written as $\boxed{v = v_{vmax} \frac{S}{S+K} M}$
- This parameters $K_M = \frac{k_1+k_2}{k_1}$ and $v_{max} = k_2\ E_{TOT}$.

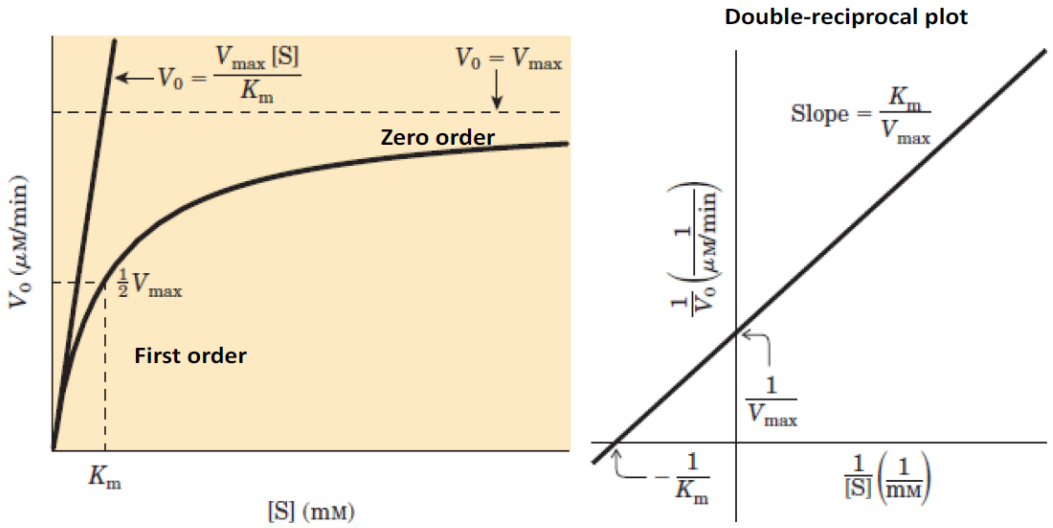
KM is the affinity and V_{max} is the maximum rate.

Lineweaver–Burk plot to estimate parameters

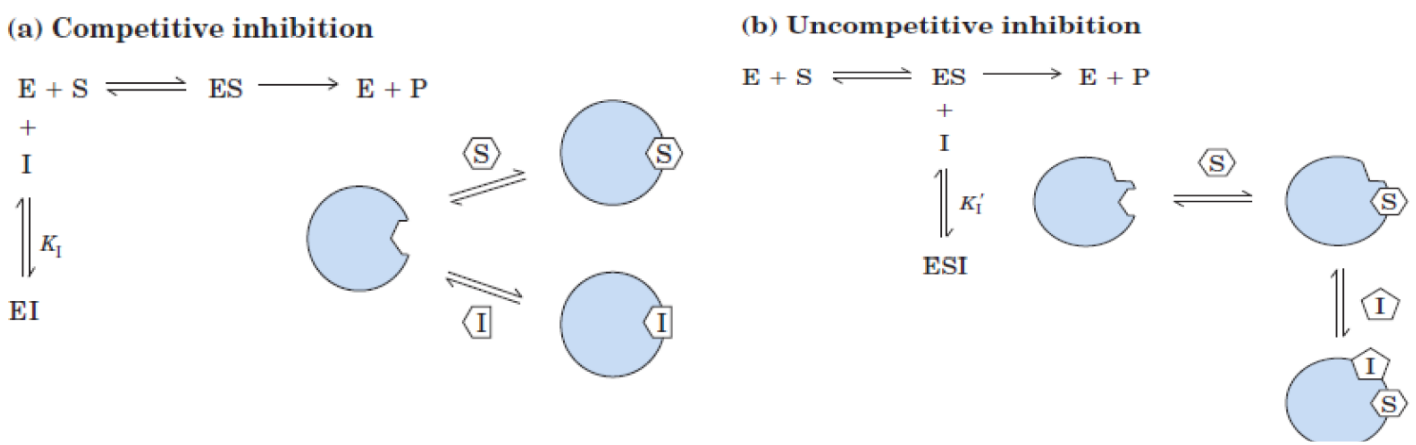
- Rewritten in the following manner, the equation gives a straight line, which is useful to determinate the parameters K_M and V_{max} (Lineweaver-Burk representation):

$$\frac{1}{v} = \frac{1}{v_{max}} + \frac{K_M}{v_{max}} \frac{1}{S}$$

- Can the two order of MM reactions be determined from elasticity ?



Lineweaver Burke plot for inhibitions



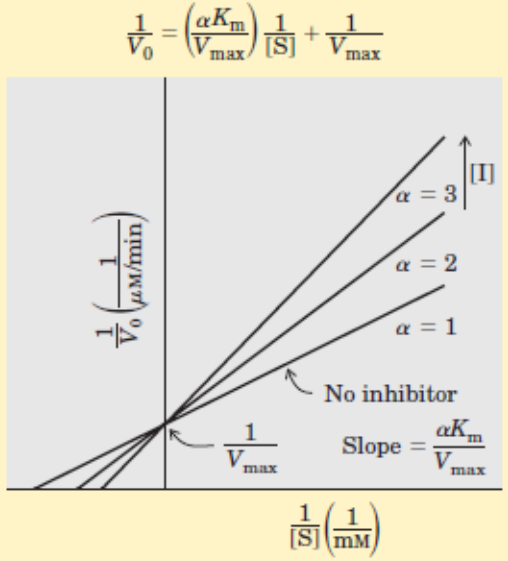


FIGURE 1 Competitive inhibition.

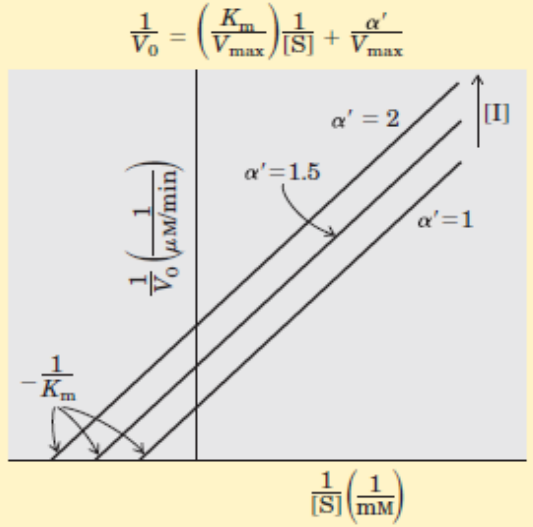
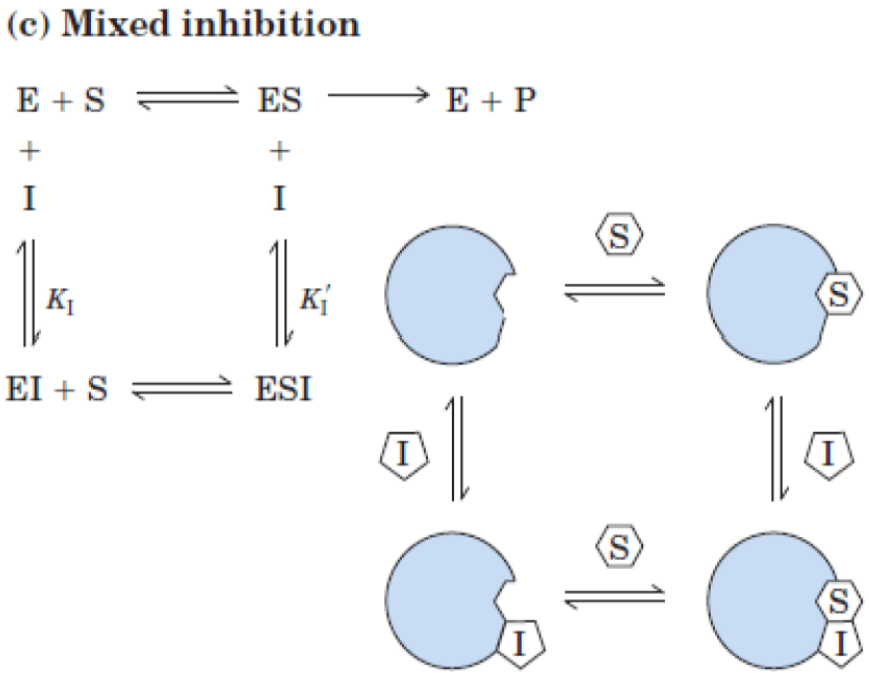


FIGURE 2 Uncompetitive inhibition.

Lineweaver Burke Plot for inhibition



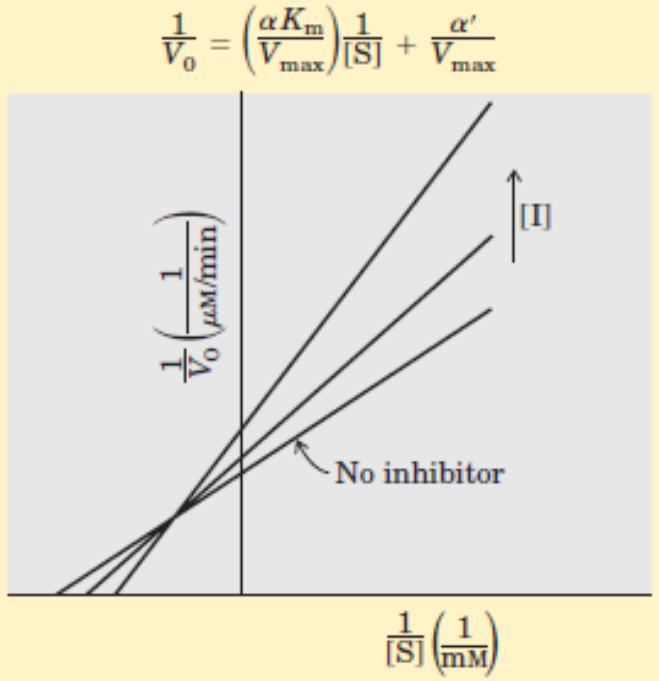


FIGURE 3 Mixed inhibition.

Steps involved in deriving rate laws in MM kinetics involving steady states

- **Step-1:** Identify the product and write the rate(s). For example, in MM kinetics, the reaction $E + S \underset{k_{-1}}{\overset{k_1}{\longleftrightarrow}} C \xrightarrow{k_2} E + P$
- Its a two step reaction. The rate of product formation is $v = k_2\,[EX]$.
- **Step-2:** Write the enzyme conservation relationship. In the above case it is $E_T = [E] + [EX]$
- **Step-3:** Write the $\frac{numerator}{denominator}$ in the form of $\frac{v}{E_T}$
- **Step-4:** All the enzyme conservation terms in Step-2 should be written in terms of free enzymes. For this complexes are written in terms of free enzyme by making steady state approximations.

Continued...

- For example, [EX] at steady state is written as k_1 [E][X] = k_2 [EX]. And [EX] = $\frac{k_1[E][X]}{k_2}$ = [EX] = $\frac{[E][X]}{k_m}$, where $\frac{k_2}{k_1}$ = k_m.

- **Step-5:** Find $\frac{V}{E_T} = \frac{k_2[EX]}{[E]+[EX]} = \frac{\frac{k_2[E][X]}{k_m}}{E+\frac{[E][X]}{k_m}}$

- **Note:** In the denominator, the free enzyme is obtained and its cancelled by the free enzyme in the numerator. So, $\frac{k_2\frac{[E][x]}{k_m}}{[E][1+frac[x]k_m]} = \frac{k_2[x]}{k_m[1+\frac{[x]}{k_m}]}$.

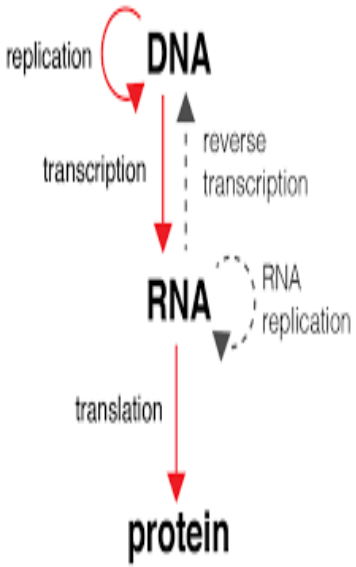
- This leads finally to $\frac{v}{E_T} = \frac{k_2[x]}{k_m+[x]}$. Therefore v = $\frac{k_2E_T[x]}{k_m+[x]}$ and if we write k_2 E_T = v_m, the final equation is $\boxed{v = \frac{v_m[x]}{k_m+[x]}}$

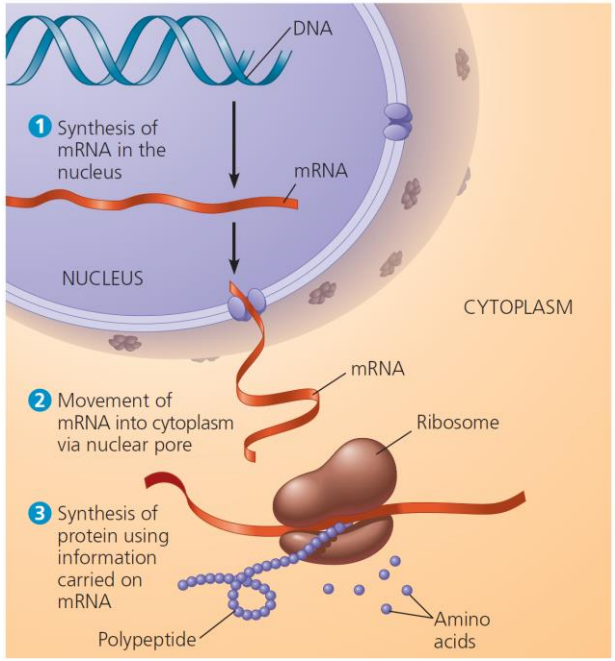
Nucleic acids–central dogma

- The amino acid sequence of a polypeptide is programmed by a discrete unit of inheritance known as a **gene.**

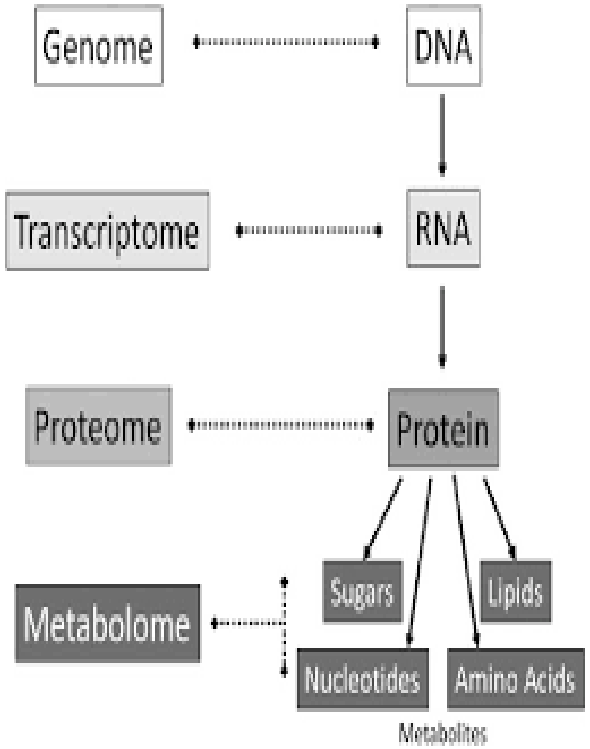
- A gene consists of DNA, a polymer known as a **nucleic acid**.
  - ✓ Nucleic acids are made of monomers called *nucleotides*.
- The two types of nucleic acids are
  - ✓ **ribonucleic acid (RNA)** and
  - ✓ **deoxyribonucleic acid (DNA)**.

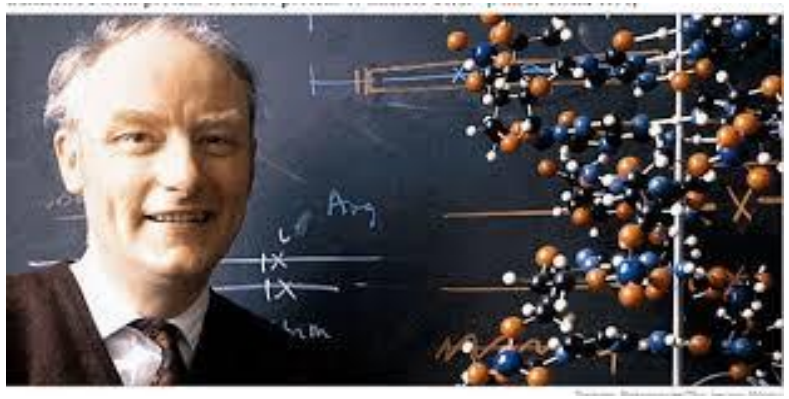
- RNA and DNA are the molecules that enable living organisms to reproduce their complex components from generation to generation.





▲ **Figure 5.23 Gene expression: DNA → RNA → protein.** In a eukaryotic cell, DNA in the nucleus programs protein production in the cytoplasm by dictating synthesis of messenger RNA (mRNA).





- The 'Central Dogma' is the process by which the instructions in DNA are converted into a functional product.

- It was first proposed in 1958 by Francis Crick, discoverer of the structure of DNA.

Nucleic acids– Nitrogeneous bases

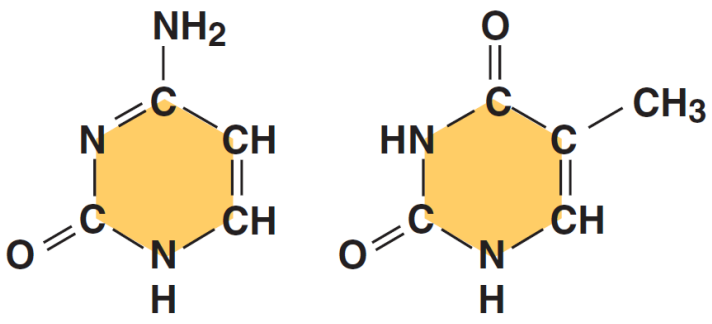
- Nucleic acids are polymers made of **nucleotide** monomers organized as **polynucleotides**.

- Each **nucleotide** consists of three parts: a nitrogenous base, a pentose sugar, and one or more phosphate groups.

- The nitrogenous bases are rings of carbon and nitrogen that come in two types: purines and pyrimidines.

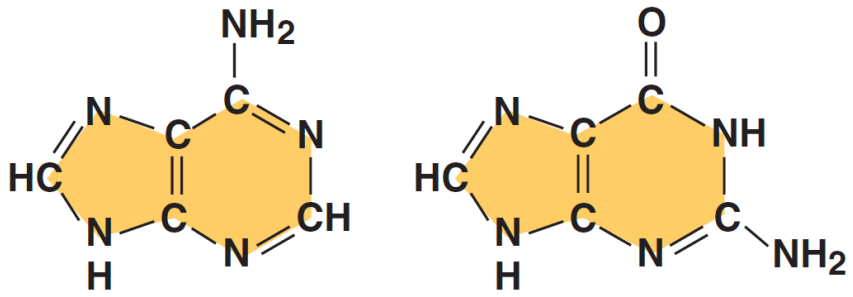
  - **Pyrimidines** have a single six-membered ring of carbon and nitrogen atoms.
    - ✓ There are three different pyrimidines: cytosine (C), thymine (T), and uracil (U).
    - ✓ Thymine is found only in DNA and uracil is found only in RNA.
  - **Purines** have a six-membered ring joined to a five-membered ring.
    - ✓ The two purines are adenine (A) and guanine (G).

Pyrimidines



Cytosine (C) **Thymine (T, in DNA)** **Uracil (U, in RNA)**

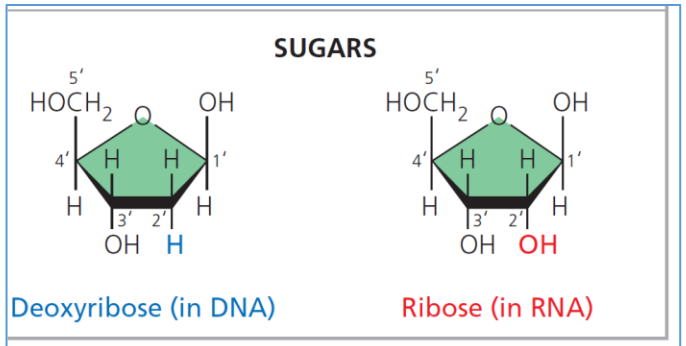
Purines



Adenine (A) **Guanine (G)**

Nucleic acid – sugar component

- The pentose joined to the nitrogenous base is **ribose** in nucleotides of RNA and **deoxyribose** in DNA.
  - ✓ *The only difference between the sugars is the lack of an oxygen atom on carbon 2 in deoxyribose.*
  - ✓ *Because the atoms in both the nitrogenous base and the sugar are numbered, the sugar atoms are distinguished by a prime (′) after the number.*
  - ✓ *Thus, the second carbon in the sugar ring is the 2′(2 prime) carbon, and the carbon that sticks up from the ring is the 5′carbon.*
  - ✓ *The combination of a pentose and a nitrogenous base is a nucleoside.*

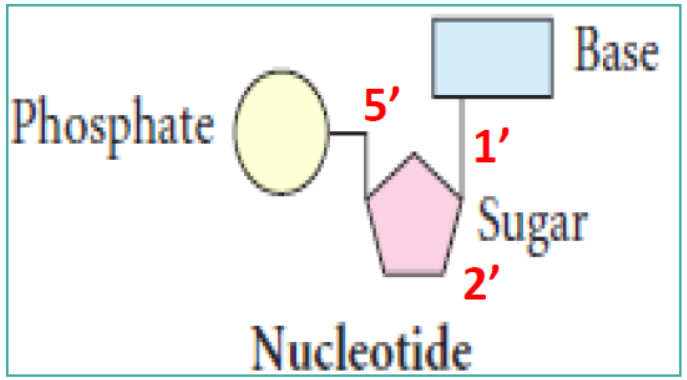


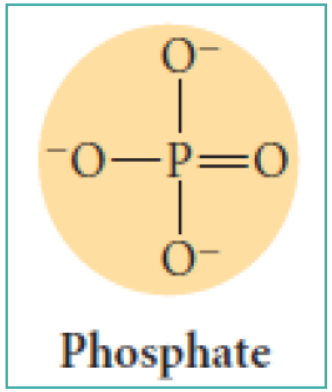
Nucleoside components

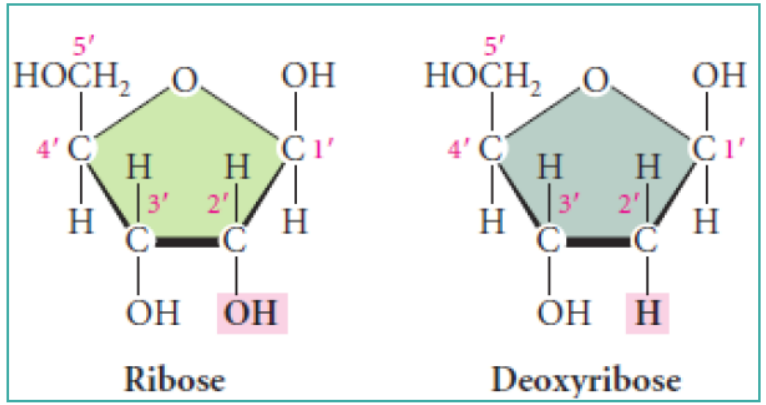
- The addition of a phosphate group creates a nucleoside monophosphate or *nucleotide.*

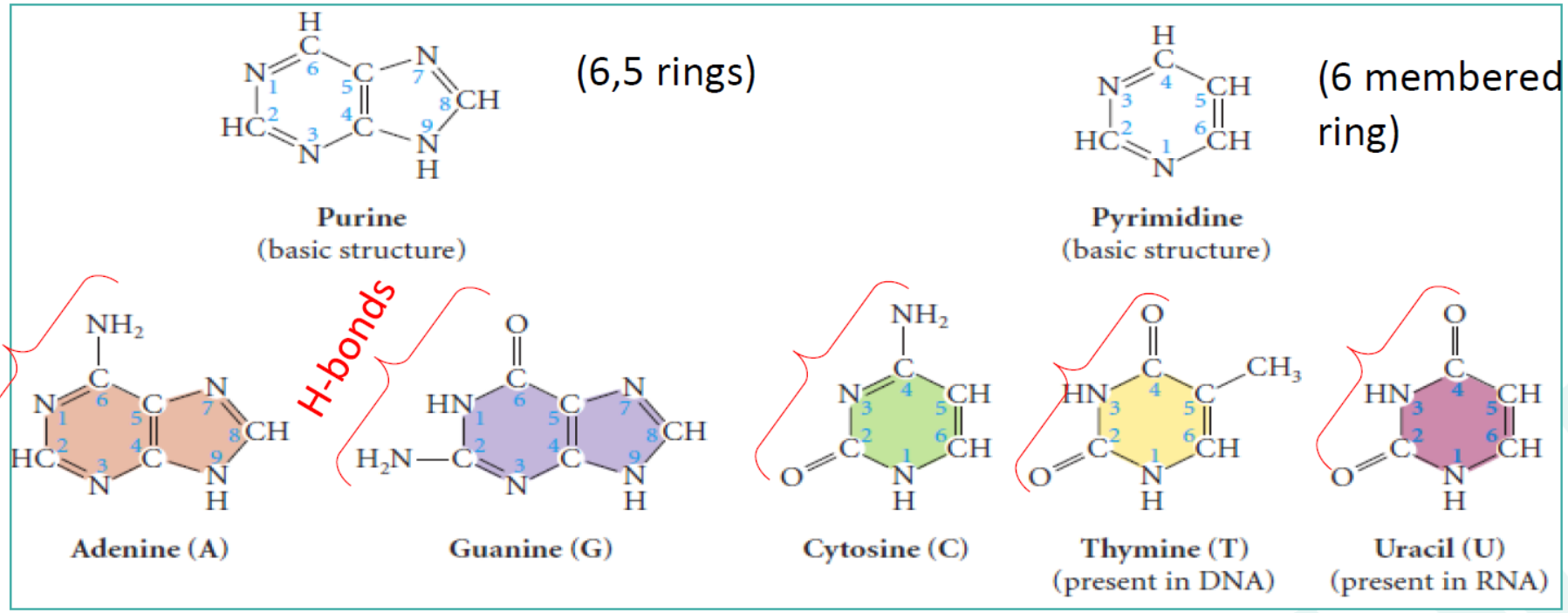
- **Polynucleotides are synthesized when adjacent nucleotides are joined by covalent bonds called phosphodiester linkages** that form between the —OH group on the 3′ of one nucleotide and the phosphate on the 5′ carbon of the next.
  - ✓ This process creates a repeating backbone of sugar-phosphate units, with appendages consisting of the nitrogenous bases.

Summary: Structure of nucleotide, nucleosides , ribose and DNA

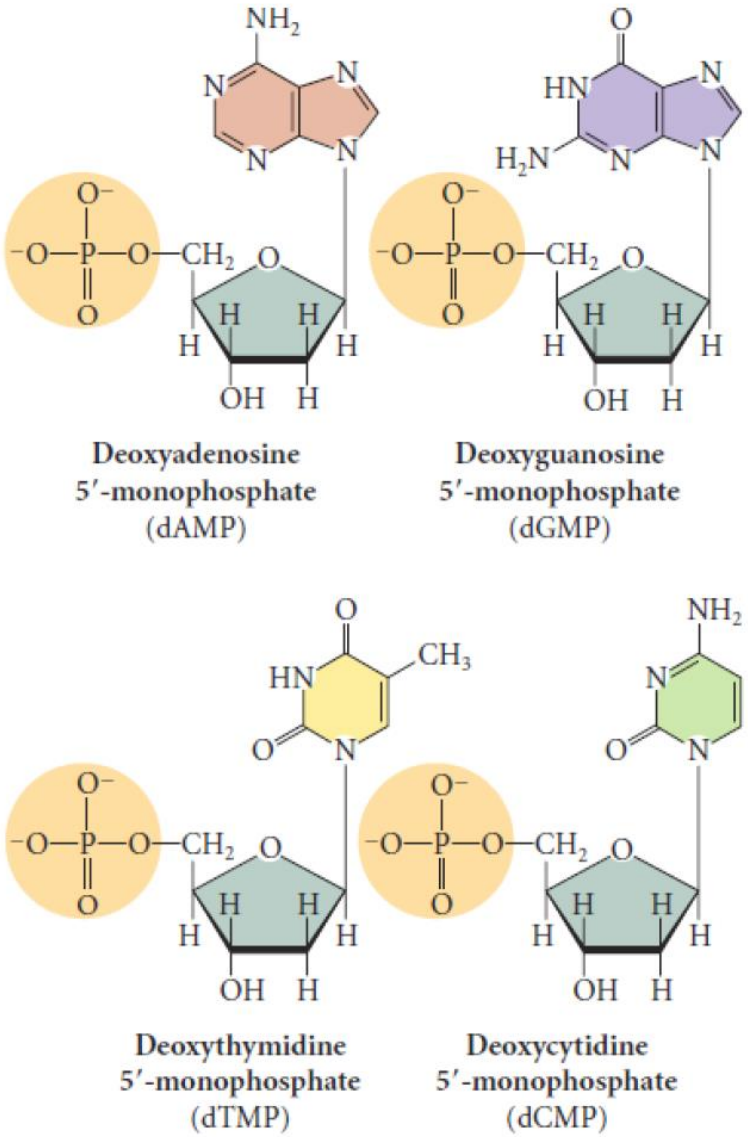








Four types of DNA nucleotides



CONCEPT CHECK

Q: How do the sugars of RNA and DNA differ?

a. RNA has a six-carbon sugar; DNA has a five-carbon sugar.

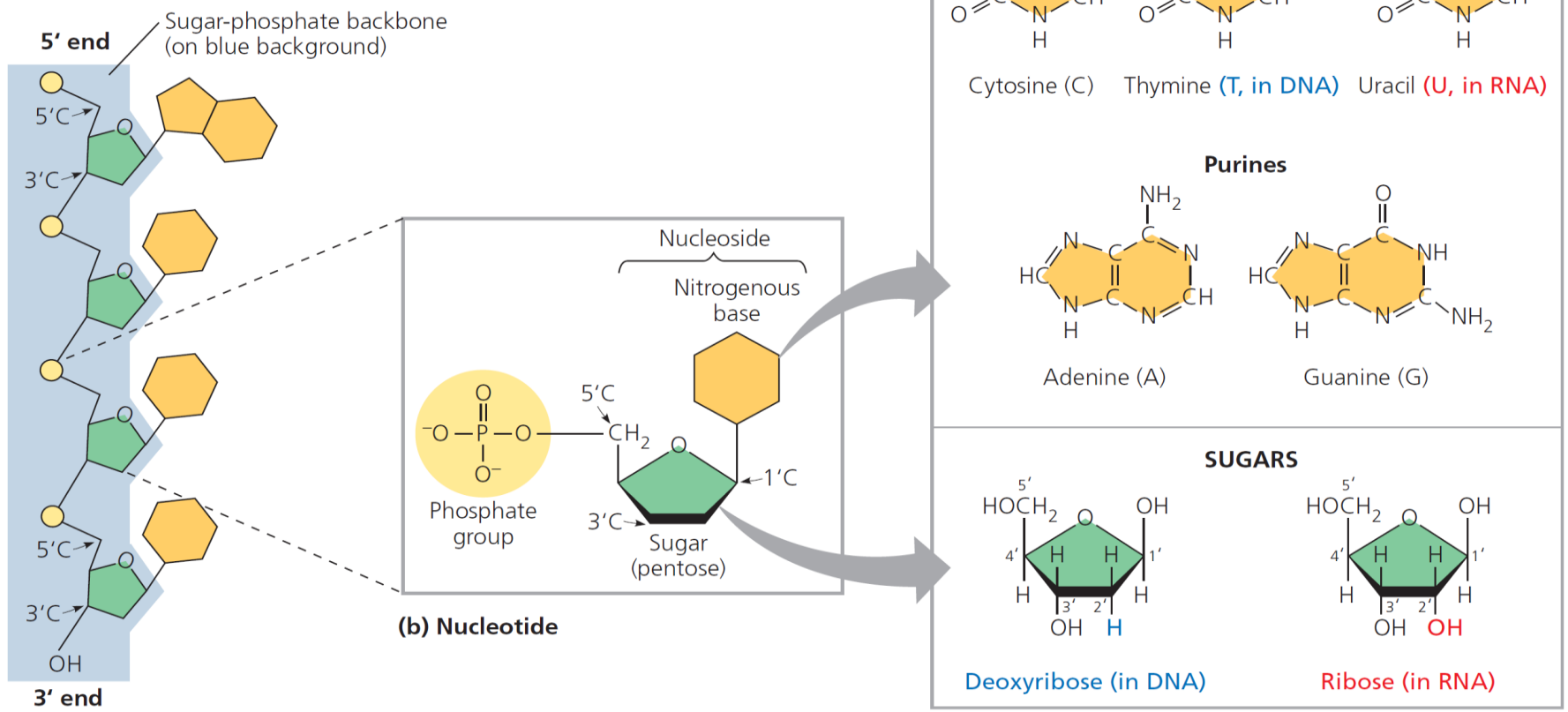
b. The sugar of RNA has a hydroxyl group that is not found in the sugar of DNA.

c. RNA contains uracil; DNA contains thymine.

d. DNA's sugar has a phosphorus atom; RNA's sugar does not.

Nucleic acids – reads from 5’ –> 3’

▼ **Figure 5.24 Components of nucleic acids. (a)** A polynucleotide has a sugar-phosphate backbone with variable appendages, the nitrogenous bases. **(b)** A nucleotide monomer includes a nitrogenous base, a sugar, and a phosphate group. Note that carbon numbers in the sugar include primes (’). **(c)** A nucleoside includes a nitrogenous base (purine or pyrimidine) and a five-carbon sugar (deoxyribose or ribose).

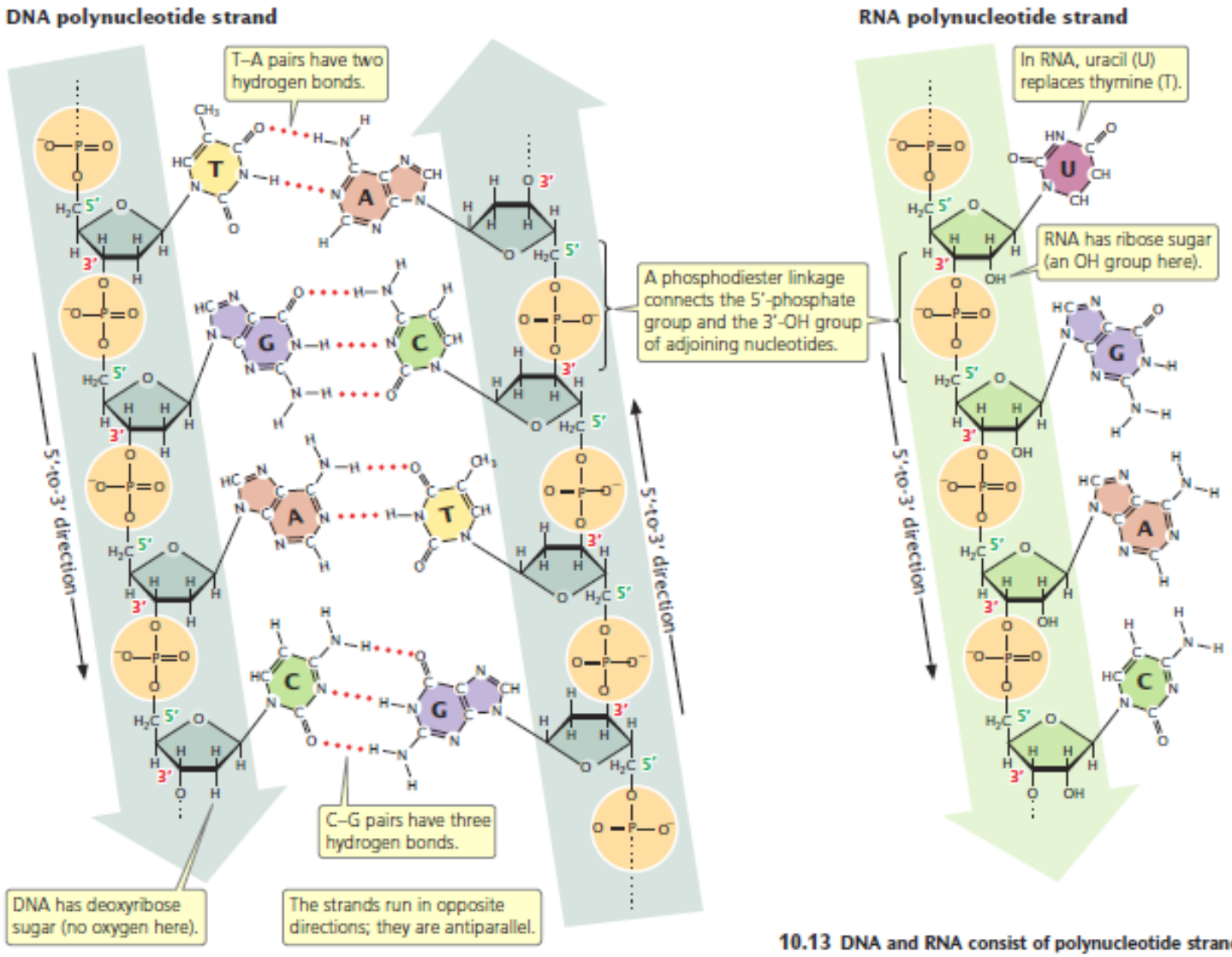


(a) Polynucleotide, or nucleic acid

(b) Nucleotide

(c) Nucleoside components

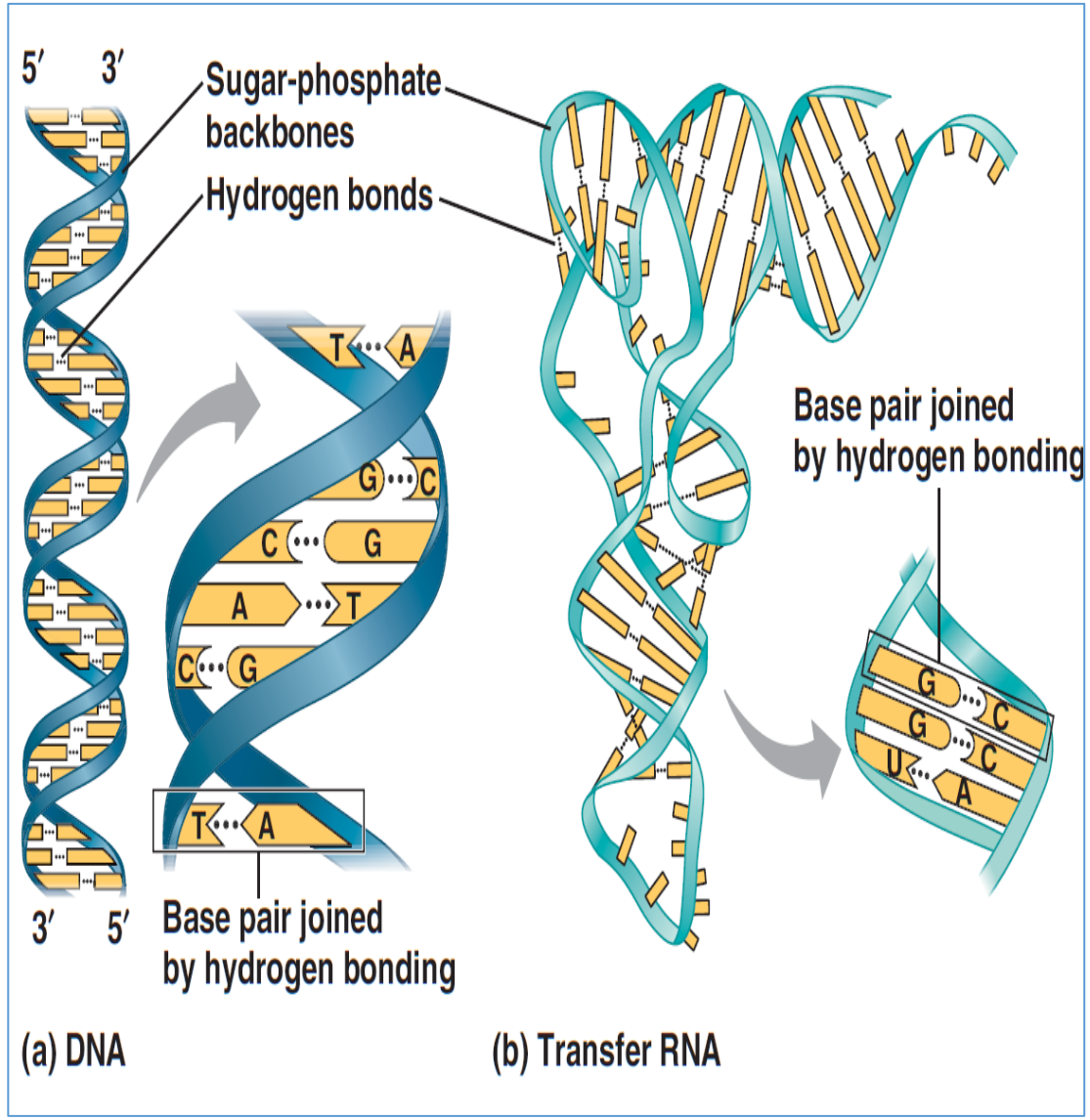
Strands in DNA—5’ is phosphate end and 3’ is OH end of sugar



10.13 DNA and RNA consist of polynucleotide strands.

DNA–hydrogen bonding

- An **RNA** molecule is **usually a single polynucleotide chain**.
- DNA molecules have two polynucleotide strands that spiral around an imaginary axis to form a **double helix**.
- The sugar-phosphate backbones of the two polynucleotides are on the outside of the helix.
  - ✓ *The two backbones run in opposite 5′ → 3′ directions from each other, an arrangement referred to as* ***antiparallel****.*
- The two polynucleotides or strands are held together by hydrogen bonds between the paired bases.
- Most DNA molecules have thousands to millions of base pairs.
- Because of their shapes, **only some bases are compatible** with each other.
  - ✓ *Adenine (A) always pairs with thymine (T) and guanine (G) with cytosine (C).*
  - ✓ *With these base-pairing rules,* ***if we know the sequence of bases on one strand, we know the sequence on the opposite strand****.*
  - ✓ *The two strands are complementary.*

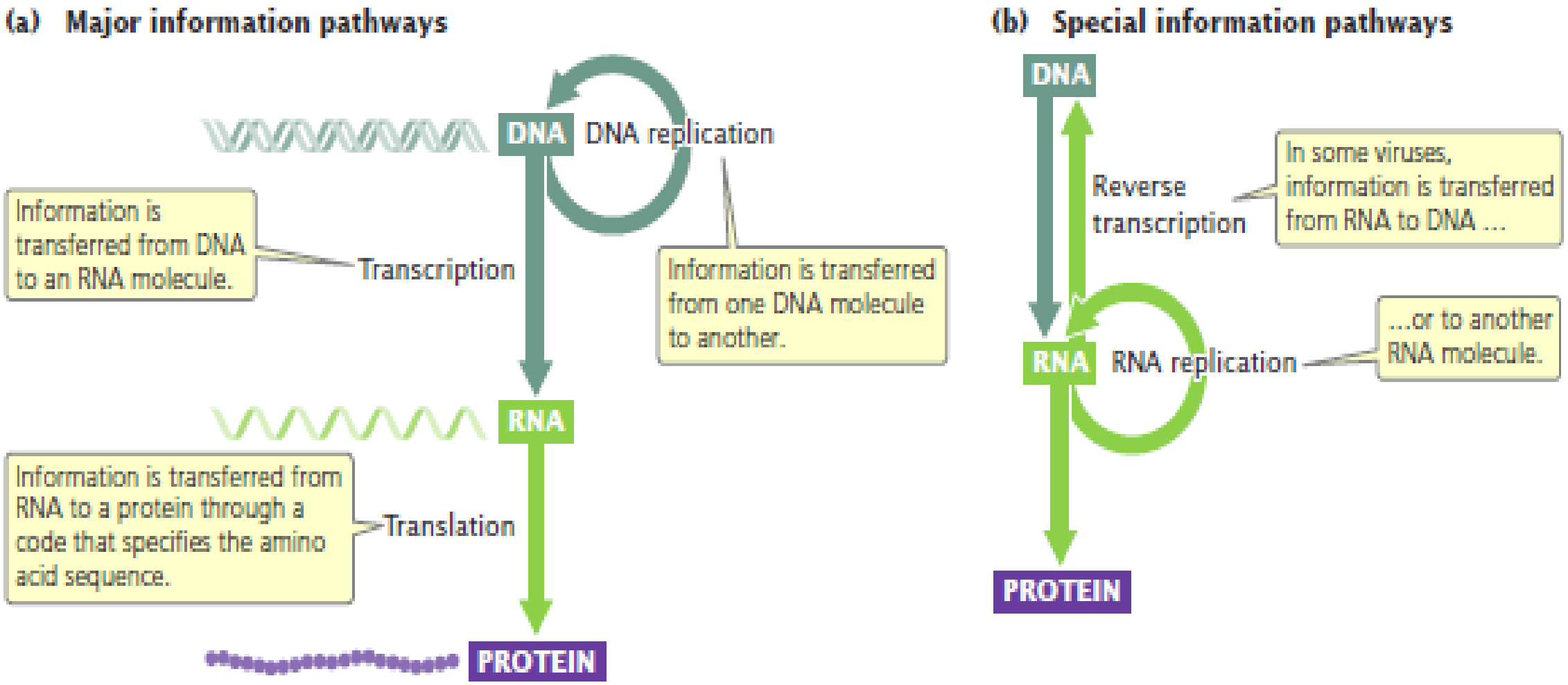


Chargaff's rule– A/T = G/C

Table 10.1 Base composition of DNA from different sources and ratios of bases

Source of DNA	A	T	G	C	Ratio A/T	Ratio G/C	Ratio (A + G)/(T + C)
E. coli	26.0	23.9	24.9	25.2	1.09	0.99	1.04
Yeast	31.3	32.9	18.7	17.1	0.95	1.09	1.00
Sea urchin	32.8	32.1	17.7	18.4	1.02	0.96	1.00
Rat	28.6	28.4	21.4	21.5	1.01	1.00	1.00
Human	30.3	30.3	19.5	19.9	1.00	0.98	0.99

Pathways of information transfer



Questions

(1) What characteristics must the genetic material possess (Give atleast two and maximum of four).

(2) Draw and identify three parts of DNA.

(3) How does RNA nucleotide differ from DNA molecule?

(4) How does purine differ from pyramidine? What purines and pyramidines are found in DNA and RNA.

(5) Draw a short segment of a single polynucleotide strand, including atleast three nucleotides. Indicate the polarity of the strand by labeling 5’ end and the 3’end.

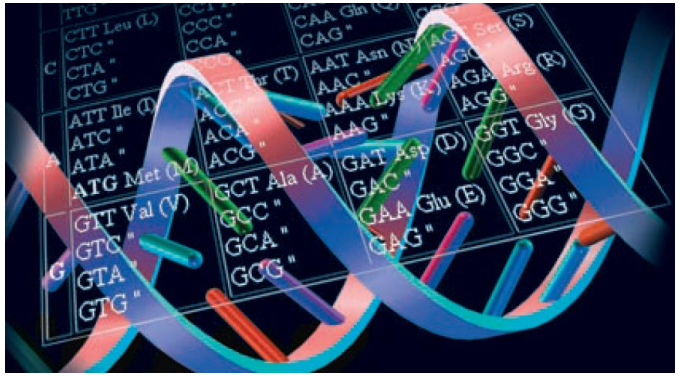
(6) What different types of chemical bonds are found in DNA and where are they found?

(7) What are the three major pathways of information that flow within cell?

(8) What are the various structures of single stranded nucleotides and how do they form?

Applications

in Science and Technology



- *Nucleotide sequencing and the analysis of large sets of genes and proteins can be done rapidly and inexpensively due to advances in technology and information processing.*
- *Taken together,* ***genomics and proteomics have advanced our understanding of biology across many different fields****.*

Paleontology

New DNA sequencing techniques have allowed decoding of minute quantities of DNA found in ancient tissues from our extinct relatives, the Neanderthals (*Homo neanderthalensis*). Sequencing the Neanderthal genome has informed our understanding of their physical appearance as well as their relationship with modern humans. *See Figure 34.49.*

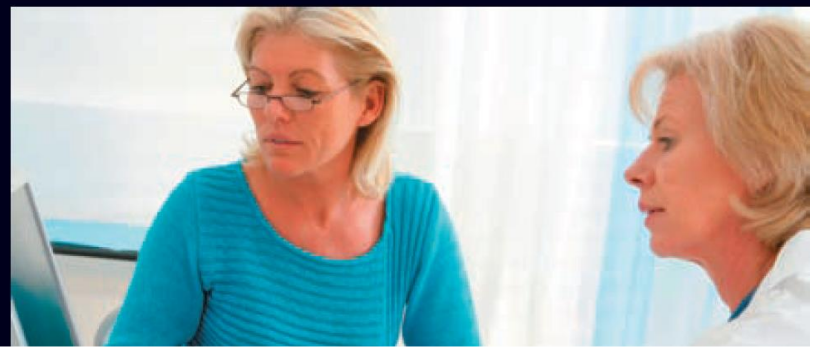
Evolution

A major aim of evolutionary biology is to understand the relationships among species, both living and extinct. For example, genome sequence comparisons have identified the hippopotamus as the land mammal sharing the most recent common ancestor with whales. *See Figure 22.20.*

Hippopotamus

Medical Science

Identifying the genetic basis for human diseases like cancer helps researchers focus their search for potential future treatments. Currently, sequencing the sets of genes expressed in an individual's tumor can allow a more targeted approach to treating the cancer, a type of "personalized medicine." *See Figures 12.20 and 18.27.*



Conservation Biology

The tools of molecular genetics and genomics are increasingly used by ecologists to identify which species of animals and plants are killed illegally. In one case, genomic sequences of DNA from illegal shipments of elephant tusks were used to track down poachers and pinpoint the territory where they were operating. *See Figure 56.9.*

Species Interactions

Over 90% of all plant species exist in a mutually beneficial partnership with fungi that are associated with the plants' roots. Genome sequencing and analysis of gene expression in several plant-fungal pairs promise major advances in our understanding of such interactions and may have implications for agricultural practices. *(See the Scientific Skills Exercise in Chapter 31.)*

MAKE CONNECTIONS *Considering the examples provided here, describe how the approaches of genomics and proteomics help us to address a variety of biological questions.*